

1. Autonomous Numeric Intelligence System

A futuristic artificial intelligence laboratory generates encrypted numeric identifiers for distributed neural synchronization systems.

During validation, the processing engine calculates both the sum and product of the digits present in the identifier. If both values are equal, the identifier is classified as a Spy Number.

The validation result is later used by synchronization controllers and predictive authentication systems.

Design a Python program using loops to determine whether a number is a Spy Number.

Input Format

Input consists of a single integer n .

Constraints

- $1 \leq n \leq 10^9$

Output Format

If the number is Spy Number, print:

Spy Number

Otherwise print:

Not Spy Number

Sample Input 0

123

Sample Output 0

Spy Number

2. Intelligent Numerical Verification Engine

A next-generation computational laboratory processes encrypted numerical identifiers generated by autonomous systems.

During validation, the engine squares the given number and calculates the sum of digits of the squared value. If the result equals the original number, the identifier is classified as a Neon Number.

Design a Python looping program to determine whether the number is Neon Number.

Sample Input

9

Sample Output

Neon Number

3. Quantum Mathematical Validation System

A futuristic quantum research organization continuously processes numerical identifiers for secure computational experiments.

During validation, the processing engine calculates the factorial of each digit and determines whether the sum equals the original number.

If the property is satisfied, the identifier is classified as a Peterson Number.

Design a Python looping program.

Sample Input

145

Sample Output

Peterson Number

4. Advanced Prime Transformation Engine

An intelligent cyber analytics organization validates secure numerical identifiers using prime transformation algorithms.

During validation, the engine reverses the number and checks whether both the original and reversed numbers are prime.

If both are prime, the identifier is classified as a Twisted Prime Number.

Design a Python program using loops.

Sample Input

13

Sample Output

Twisted Prime

5. Secure Reverse Prime Verification System

A distributed cyber security organization processes encrypted numeric identifiers for authentication systems.

During analysis, the validation engine checks whether a number is prime and whether its reverse is a different prime number.

If both conditions are satisfied, the number is classified as an Emirp Number.

Design a Python program using loops.

Sample Input

17

Sample Output

Emirp Number

6. Intelligent Numeric Pattern Analyzer

A futuristic communication laboratory validates encrypted transmission identifiers generated by autonomous devices.

During processing, the system determines whether the number contains zero digits except at the beginning of the number.

If the property is satisfied, the identifier is classified as a Duck Number.

Design a Python looping program.

Sample Input

1205

Sample Output

Duck Number

7. Digital Publication Authentication Engine

A multinational digital publishing organization processes ISBN identifiers for intelligent document validation systems.

During verification, the engine calculates the weighted sum of digits and checks whether the identifier satisfies ISBN validation conditions.

Design a Python looping program to validate a 10-digit ISBN number.

Sample Input

020131452

Sample Output

Valid ISBN

8. Neural Binary Distance Analyzer

An advanced robotics laboratory processes binary synchronization sequences for distributed neural communication systems.

During analysis, the validation engine must determine the maximum distance between two consecutive 1s in the binary representation of a number.

Design a Python looping program.

Sample Input

22

Sample Output

2

Explanation

Binary representation:

10110

Maximum gap:

2

9. Autonomous Numeric Compression System

A next-generation AI compression engine continuously processes large numerical identifiers.

During optimization, the engine repeatedly calculates the sum of digits until a single digit is obtained. The final digit is called the Digital Root.

Design a Python program using loops.

Sample Input

9876

Sample Output

3

10. Intelligent Prime Rotation Engine

A futuristic computational research organization validates encrypted prime identifiers used by distributed autonomous systems.

During validation, the engine rotates all digits of the number and checks whether every rotation forms a prime number.

If all rotations are prime, the identifier is classified as a Circular Prime.

Design a Python looping program.

Sample Input

197

Sample Output

Circular Prime